Homework: Bootstrap, CLT, and Normal Approximation

1. Random sample has: $\bar{X}=52$, $S=10$, $n=100$

Compute estimated SE of $\bar{X}$. Tests $H_0: \mu=50$ $H_1: \mu \neq 50$ at significance levels $\alpha = .10, .05, .01$ using large sample z-statistic. Construct a 95% CI for μ.

$\widehat{SE}(\bar{X}) = \frac{S}{\sqrt{n}} = \frac{10}{\sqrt{100}} = 1$

$z = \frac{\bar{X}-\mu_0}{\widehat{SE}} = \frac{52-50}{1} = \frac{2}{1} = 2$ (52 is 2 SEs above 50)

α	Critval	Is $\lvert z \rvert$ bigger?	
0.10	1.645	Yes	REJECT H_0
0.05	1.96	Yes	REJECT H_0
0.01	2.576	No	FAIL to REJECT H_0

$\bar{X} \pm 2 \cdot \widehat{SE} = [52 + 1.96(1), 52 + 1.96(1)] = [50.04, 53.96]$ 95% CI

2. $\hat{S} = 3.1$, $\widehat{SE}(\hat{S}) = 1.3$ $H_0: S=0$ $H_1: S \neq 0$

$z = \frac{3.1-0}{1.3} = 2.385$

90% CI: $[3.1 - 1.645(1.3), 3.1 + 1.645(1.3)] = [0.962, 5.239]$

95% CI: $[3.1 - 1.96(1.3), 3.1 + 1.96(1.3)] = [0.552, 5.648]$

99% CI: $[3.1 - 2.576(1.3), 3.1 + 2.576(1.3)] = [-0.249, 6.449]$

Fail to reject the null at 99% confidence, reject the null at 90% and 95% confidence.

Health survey compares a biomarker to a benchmark value of 12.

$\bar{X} = 11.5$, $S=3$, $n=36$

Is 12 a plausible value for the pop. mean? Construct a 95% CI and compute 2-sided p-value for $H_0: \mu = 12$ versus $H_1: \mu \neq 12$. Assess plausibility using interval and test at $\alpha = 0.05$.

$z = \frac{11.5-12}{.5} = \frac{-.5}{.5} = -1$ $\widehat{SE} = \frac{3}{\sqrt{36}} = \frac{1}{2}$

95% CI: $[11.5 - (\frac{1}{2})(1.96), 11.5 + (\frac{1}{2})(1.96)] = [10.52, 12.48]$

$p = 2\Phi(-|z|)$ $2 \cdot 0.1587 = 0.3173 > 0.05$ Fail to reject $H_0: \mu = 12$.

↑ 2-sided test

12 is a plausible value - it's in the 95% CI and the z-score of -1 shows 12 w/in 1 SE of sample mean.

4. A real estate analyst compares the avg price of a house in 2 neighborhoods

Group A: $\bar{X}_A = 25$, $s_A = 8$, $n_A = 100$

Group B: $\bar{X}_B = 22$, $s_B = 7$, $n_B = 100$

Compute 2-sample z-stat for equality of means. Construct a 95% CI for $\mu_A - \mu_B$

$$z = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

$SE_A = \frac{8}{10} = 0.8$

$SE_B = \frac{7}{10} = 0.7$

$$Z = \frac{25 - 22 - (0)}{\sqrt{\frac{8^2}{100} + \frac{7^2}{100}}} = \frac{3}{1.063} \approx 2.82$$

95% CI: $3 \pm 1.96(1.063) = [0.92, 5.08]$

Interval excludes 0, difference is statistically significant at 0.05

5. Treatment A: $\bar{X}_A = 14$, $s_A = 6$, $n_A = 81$

Treatment B: $\bar{X}_B = 12$, $s_B = 5$, $n_B = 64$

Compute SE of difference in means and corresponding test statistic for

$H_0: \mu_A - \mu_B = 0$

$SE_A = \frac{6}{9}$

$SE_B = \frac{5}{8}$

$(SE_A - SE_B) = \sqrt{\frac{6^2}{81} + \frac{5^2}{64}} = 0.914$

$z = \frac{2 - 0}{.914} \approx 2.19$

$|z| > 1.645$ and 1.96 but not 2.576

reject H_0 at $\alpha = 0.10$ and 0.05, fail at $\alpha = 0.01$

6. Factory A: $\bar{X}_A = 3.2$, $s_A = 1.5$, $n_A = 49$

Factory B: $\bar{X}_B = 2.8$, $s_B = 1.4$, $n_B = 64$

Is diff statistically large relative to noise? Compute test statistic and 2-sided p-value for $H_0: \mu_A - \mu_B = 0$ and interpret result at $\alpha = 0.05$

$(SE_A - SE_B) = \sqrt{\frac{1.5^2}{49} + \frac{1.4^2}{64}} \approx 0.278$

$z = \frac{0.4 - 0}{.278} = 1.446$

$p = 2 \cdot \Phi(-|z|) = 0.1482$ - fail to reject $H_0: \mu_A = \mu_B$ at 0.05 significance level